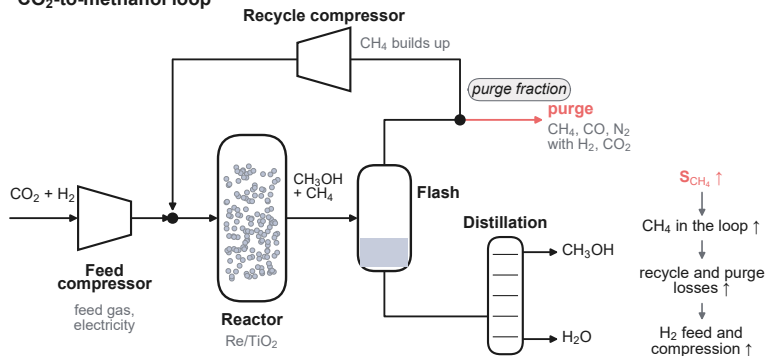
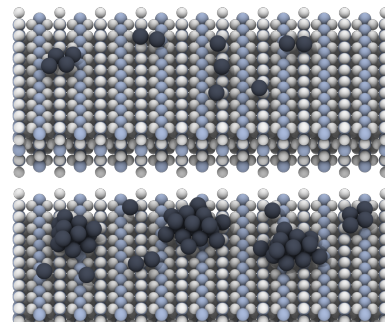


a**CO₂-to-methanol loop****b****Re/TiO₂ catalyst states (schematic)**

measured at 200 and 250 °C, 100 bar,
CO₂ : H₂ = 1 : 4

1 wt% Re
12 Re atoms

5 wt% Re
60 Re atoms

**c****Catalyst, measured**

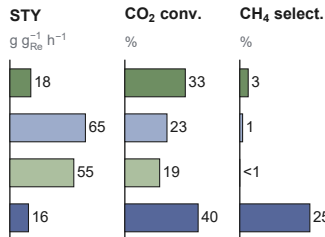
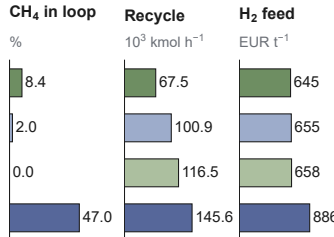
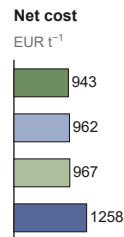
Catalyst state
in net-cost order

■ 5 wt%, 200 °C

■ 1 wt%, 250 °C

■ 1 wt%, 200 °C

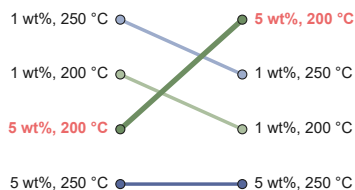
■ 5 wt%, 250 °C

**Loop at 2% purge****Plant****d Rank probability**

5,000 measurement draws

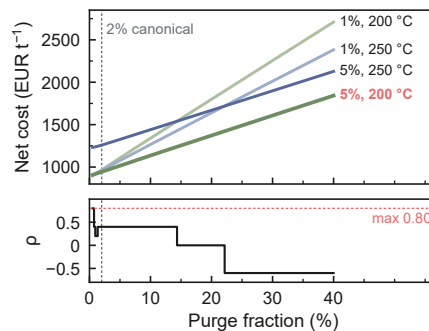
#1	#2	#3	#4
0.91	0.08	0.01	
0.08	0.71	0.21	
0.01	0.21	0.78	
			1.00

#1 kept in 4,559 / 5,000
#2 ↔ #3 swap in 1,054 / 5,000

e**STY per g Re Net cost**

$\rho = 0.40$, $\tau = 0.33$

2 of 6 pairs inverted

f**g**